Random Variables and Probability Distributions

Code

STAT 4714

AUTHOR
Michael Schwob

1 Definitions and Classifications

1.1 Experiments & Random Variables

A **statistical experiment** is used to describe any process by which several chance observations are generated. It is often important to allocate a numerical description to the outcome, which we call a **random variable**. A random variable is a function that associates a real number with each element in the sample space.

1.2 Experiments & Random Variables

- Ex: Suppose we test 3 electronic components to see if they are defective. The sample space is

$$S = \{NNN, NND, NDN, NDD, DNN, DND, DDN, DDD\}.$$

  If we are interested in the number of defectives that occur, our random variable X would denote the numerical values 0, 1, 2, or 3 assigned to each event. Each possible value of X represents an event that is a subset of the sample space for the given experiment.

1.3 Random Variables and Notation

- Ex: In the previous example with defective electronic components, note that $X = 2$ for the following elements:

$$E = \{NDD, DND, DDN\}.$$

- *A note on notation:* We will use the capital letter X to denote a random variable and a lowercase letter x to denote a corresponding value for the random variable.

- Ex: For the outcome DDN, we say $x = 2$. However, $X = 2$ suggests an event that contains several elements, one of them being DDN.

1.4 Ex: The Common Urn

Two balls are drawn in succession without replacement from an urn containing 4 red balls and 3 black balls. Let Y denote the number of red balls. The possible outcomes and the values y of the random variable Y are

Sample Space	$\boldsymbol{y}$
RR	2
RB	1
BR	1
BB	0

1.5 Ex: Binary Outcomes

Consider the simple condition in which components are arriving from the production line and they are stipulated to be defective or non-defective. Define the random variable X by

$$X = \begin{cases} 1, & \text{if the component is defective,} \\ 0, & \text{if the component is not defective.} \end{cases}$$

Note that the assignment of whether defectives or non-defectives get "1" is arbitrary. It would be virtually the same if

$$X = \begin{cases} 0, & \text{if the component is defective,} \\ 1, & \text{if the component is not defective.} \end{cases}$$

1.6 Discrete Sample Spaces

A sample space is **discrete** if it contains a finite number of possibilities or an unending sequence with as many elements as there are whole numbers.

- Ex: Suppose a circuit has 3 fuses, each of which are either working (W) or not working (N). The sample space is discrete with $2^3 = 8$ outcomes.

- Ex: We are interested in packet transmission in a network. Each transmitted packet can be classified as success (S), corrupted (C), or lost (L). For two packets that are sent, there are $3^2 = 9$ outcomes, so S is discrete.

1.7 Continuous Sample Space

A sample space is **continuous** if it contains an infinite number of possibilities equal to the number of points on a line segment.

- Ex: The voltage amplitude of a received analog signal can take any value in a range between $[-5V, +5V]$, so $S = [-5, 5]$ is continuous.
- Ex: The lifetime of a resistor (in hours) can be modeled as any non-negative real number. Then, $S = [0, \infty)$. This is an unbounded continuous sample space, often modeled with an exponential or Weibull distribution.

1.8 Random Variables

- If a random variable has a countable set of possible outcomes, it is a **discrete random variable**. If it can take on values in a continuous scale, it is a **continuous random variable**.
- In most applications, continuous random variables represent measured data, whereas discrete random variables represent count data (or indicators).
- *Note for future you:* An easy way to determine appropriate probability distributions for your problem is to look at the support of the data.

2 Discrete Probability Distributions

2.1 Probability Mass Functions

It is often convenient to represent all the probabilities of a random variable X by a formula that is a function of the numerical values that X can take on. The set of ordered pairs $(x, f(x))$ is a probability function, **probability mass function**, or probability distribution of the discrete random variable X if, for each possible outcome x,

1. $f(x) \geq 0$,
2. $\sum_x f(x) = 1$,

where $f(x) = P(X = x)$.

2.2 Ex: Bit Errors in Transmission

A transmitter sends $n = 8$ bits across a noisy channel. Each bit has an independent probability of being flipped, resulting in an error: $p = 0.01$. Let X denote the number of bit errors in 8 transmissions. The probability mass

function (pmf) of X is

$$f(x) = P(X = x) = \binom{8}{x}(0.01)^x(0.99)^{8-x}$$

for $x = 0, 1, \ldots, 8$. This pmf is known as the **binomial distribution**, providing the exact probability of observing exactly x bit errors.

2.3 Ex: Packet Arrivals

Packets arrive at a router with rate $\lambda = 3$ packets per millisecond. Let Y denote the number of packet arrivals in one millisecond. The pmf of Y is

$$f(y) = \frac{e^{-\lambda}\lambda^y}{y!}$$

for $y = 0, 1, 2, \ldots$. This pmf is known as the **Poisson distribution** and is often used when dealing with *rates* or *per-area* problems.

2.4 Cumulative Probabilities

In many problems, we may be interested in computing the probability that the observed value of X will be less than or equal to some real number (i.e., the cumulative probability). In such cases, we would use the **cumulative distribution function** $F(x)$.

For a discrete random variable X with pmf $f(x)$, the cumulative distribution function (cdf) is

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t)$$

for $-\infty < x < \infty$.

2.5 Ex: Bit Errors and Packets

- *Bit Errors:* The CDF for X, where $x = 0, 1, 2, \ldots, 8$, is

$$F_X(x) = P(X \leq x) = \sum_{i=0}^{x} P(X = i) = \sum_{i=0}^{x} \binom{8}{i}(0.01)^i(0.99)^{8-i}$$

- *Packet Arrivals:* The CDF for Y is

$$F_Y(y) = P(Y \leq y) = \sum_{i=0}^{y} P(Y = i) = \sum_{i=0}^{y} \frac{e^{-\lambda}\lambda^i}{i!}$$

  for $y = 0, 1, 2, \ldots$.

2.6 PMF vs CDF

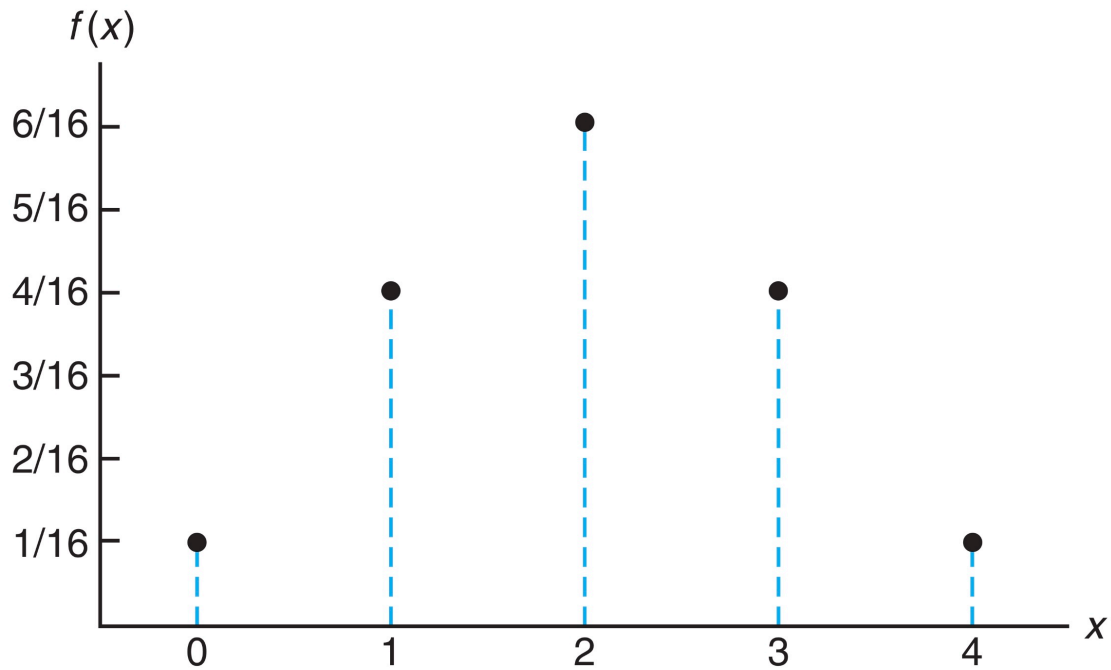


Figure 3.1: Probability mass function plot.

2.7 PMF vs CDF

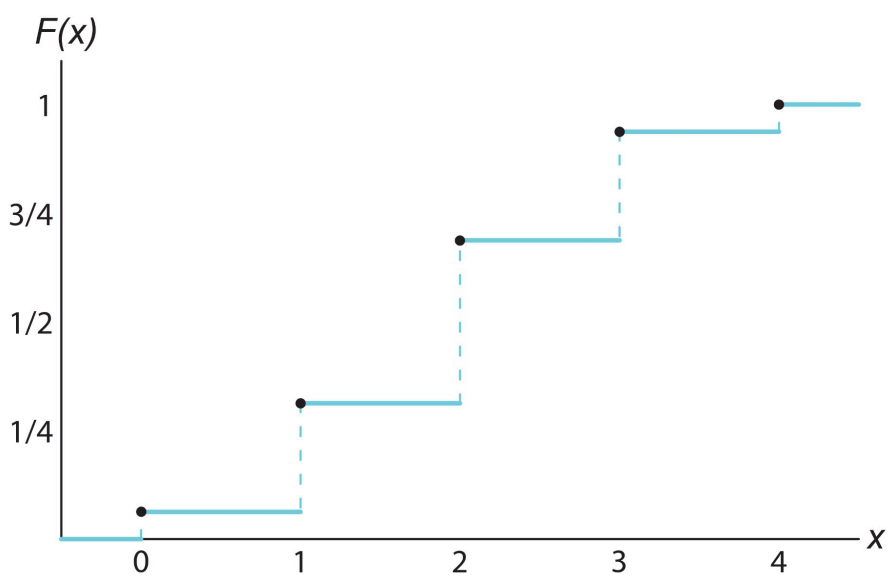


Figure 3.3: Discrete cumulative distribution function.

2.8 Breakout Groups

See Ch3-BO1

3 Continuous Probability Distributions

3.1 Probability Density Functions

The continuous analog to the probability mass function is the **probability density function**. The change in name emphasizes the concept that the probability of a continuous random variable taking any exact value is 0. There is no mass at any one particular value; rather, we concern ourselves with computing probabilities for various intervals of continuous random variables, such as $P(a < X < b)$ or $P(Y \geq c)$.

Note that when X is continuous,

$$P(X \leq b) = P(X < b) + P(X = b) = P(X < b),$$

so we can include or exclude endpoints however we wish.

3.2 Probability Density Functions

The function $f(x)$ is a **probability density function** (pdf) for the continuous random variable X, defined over the set of real numbers, if

1. $f(x) \geq 0$ for all $x \in \mathbb{R}$,
2. $\int_{-\infty}^{\infty} f(x)\, dx = 1$,
3. $P(a < X < b) = \int_a^b f(x)\, dx$.

3.3 Probabilities of Continuous Intervals

$$P(a < X < b) = \int_a^b f(x)\ dx.$$

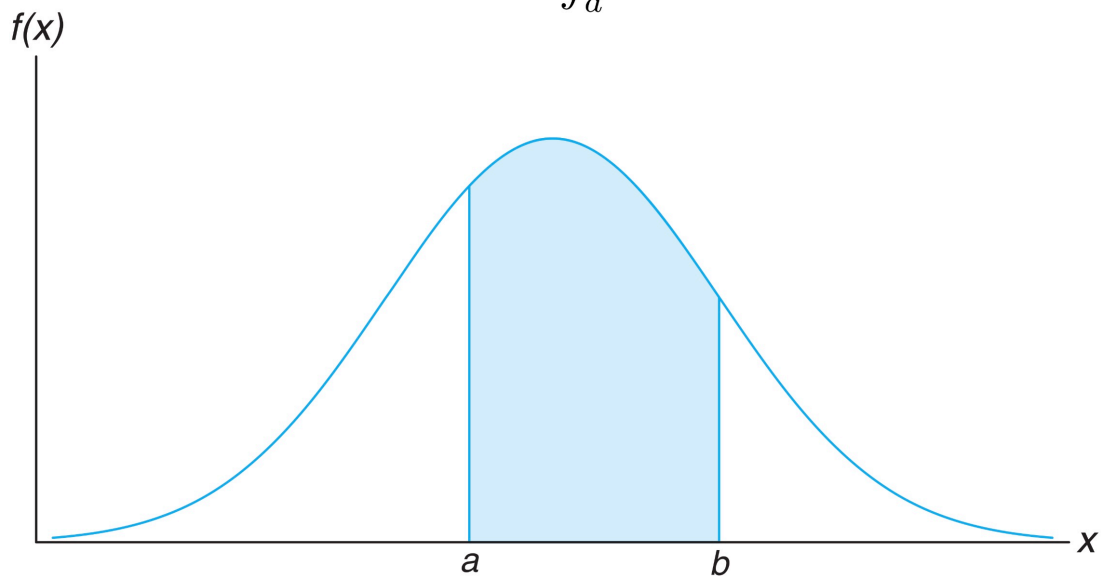


Figure 3.5: $P(a < X < b)$.

3.4 Ex: Noisy Communication Channel

Thermal/electronic noise in resistors or communication channels is often modeled as Gaussian (normal). Let X denote the noise in a communication channel. The pdf of X is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)$$

for $-\infty < x < \infty$. In ECE, this is known as the **additive white Gaussian noise (AWGN) model**, which is used frequently in digital communications analysis.

3.5 Ex: Lifetime of a Capacitor

The time-to-failure of a capacitor, transistor, or resistor under constant stress is often modeled with the **exponential distribution**. Let T denote the time-to-failure of a capacitor. The pdf of T is

$$f_T(t) = \lambda e^{-\lambda t}$$

for $t \geq 0$. In ECE, the exponential distribution is often used in reliability analysis and queueing systems.

3.6 Cumulative Probabilities

The cumulative distribution function $F(x)$ of a continuous random variable X with density function $f(x)$ is

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\, dt$$

for $-\infty < x < \infty$.

As a consequence of this definition, note that

$$P(a < X < b) = F(b) - F(a) \quad \text{and} \quad f(x) = \frac{dF(x)}{dx}.$$

3.7 A CDF Visual

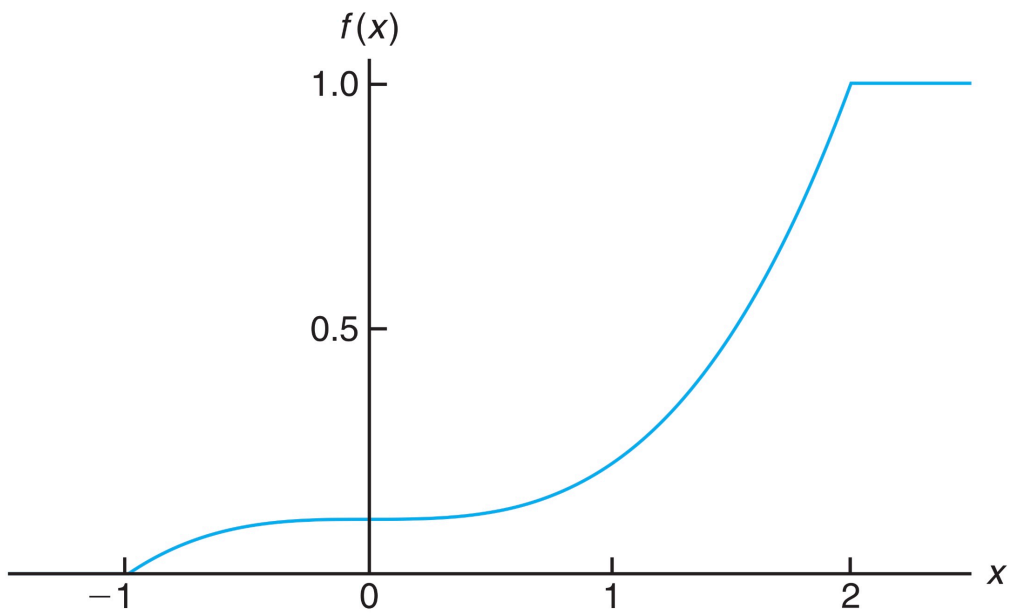


Figure 3.6: Continuous cumulative distribution function.

3.8 Ex: Noisy Communication & Time-to-Failure

- *Noisy communication channel:* The CDF of X is

$$F_X(x) = P(X \leq x) = \frac{1}{2}\left[1 + \text{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right]$$

  for $-\infty < x < \infty$, where erf$(\cdot)$ is the error function.

- *Lifetime of a capacitor:* The CDF of T for $t \geq 0$ is

$$F_T(t) = P(T \leq t) = 1 - e^{-\lambda t}.$$

3.9 Breakout Groups

See Ch3-BO2

4 Joint Probability Distributions

4.1 Simultaneous Outcomes

In many real-world problems, we are interested in studying simultaneous outcomes of several random variables.

- Ex: In digital communications, the received signal is often modeled as two orthogonal components: (I, Q), where I is the in-phase component and Q is the quadrature component.
- Ex: Suppose we measure the random voltage V across a resistor and the current I through it under noisy conditions.
- Ex: For a router, we may track the inter-arrival time between packets and the service time required by the router.

4.2 Joint Probability Distribution

The function $f(x, y)$ is a **joint probability distribution** or **joint probability mass function** of the discrete random variables X and Y if

1. $f(x, y) \geq 0$ for all (x, y),
2. $\sum_x \sum_y f(x, y) = 1$,
3. $P(X = x, Y = y) = f(x, y)$.

Note that for any region A in the xy plane,

$$P[(X, Y) \in A] = \sum \sum_A f(x, y).$$

4.3 Ex: Ballpoint Pens

Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If X is the number of blue pens selected and Y is the number of red pens selected, find the joint probability function $f(x, y)$.

- **Let's start with $f(0, 1)$. There are $\binom{8}{2} = 28$ ways to select 2 pens. The number of ways to select 0 blue pens and 1 red pen is $\binom{2}{1}\binom{3}{1} = 6$ ways. So, $f(0, 1) = 6/28$. We can repeat this for each possible set of inputs...**

4.4 Ex: Ballpoint Pens (table)

	$f(x,y)$	x = 0	1	2	Row Totals
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

4.5 Ex: Ballpoint Pens (region probability)

Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If X is the number of blue pens selected and Y is the number of red pens selected, find $P[(X,Y) \in A]$, where A is the region $\{(x,y) \mid x+y \leq 1\}$.

- **Just add the appropriate joint outcomes:**

$$\begin{aligned} P[(X,Y) \in A] &= P(X+Y \leq 1) \\ &= f(0,0) + f(0,1) + f(1,0) = 9/14. \end{aligned}$$

4.6 Joint Density Function

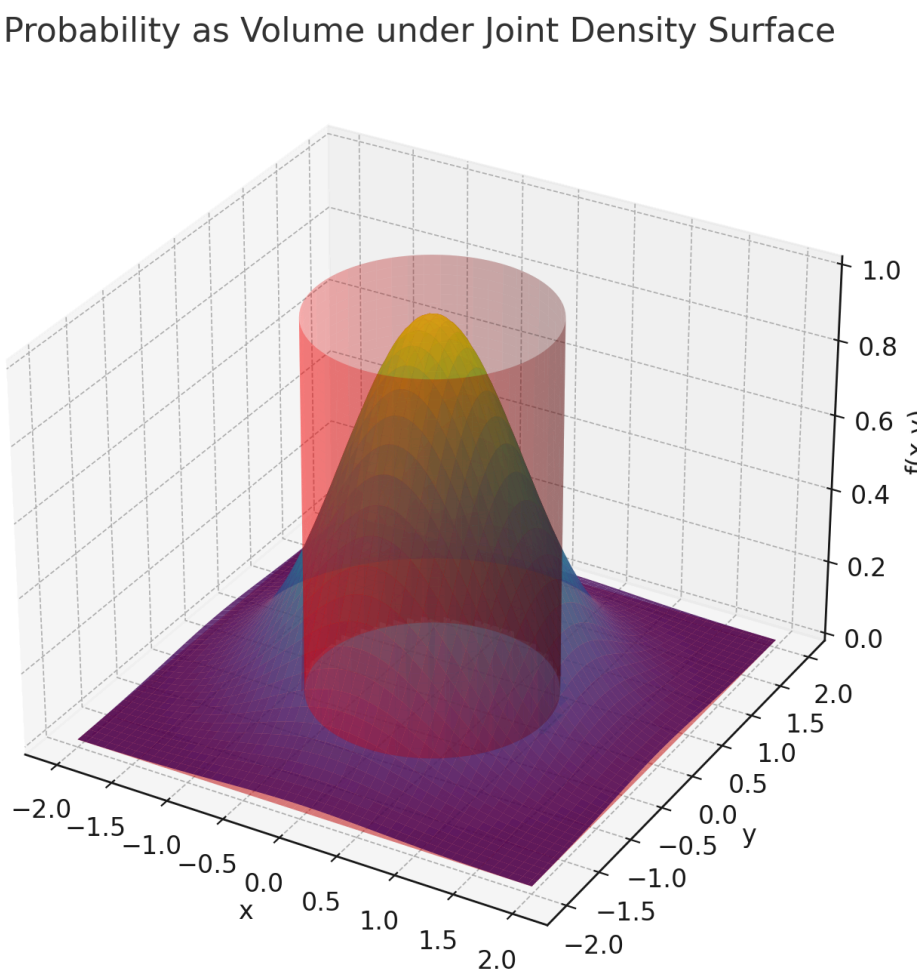
Formally: The function $f(x,y)$ is a **joint density function** of the continuous random variables X and Y if

1. $f(x,y) \geq 0$ for all (x,y),
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\, dx\, dy = 1$,
3. $P[(X,Y) \in A] = \int\int_A f(x,y)\, dx\, dy$ for any region A in the xy plane.

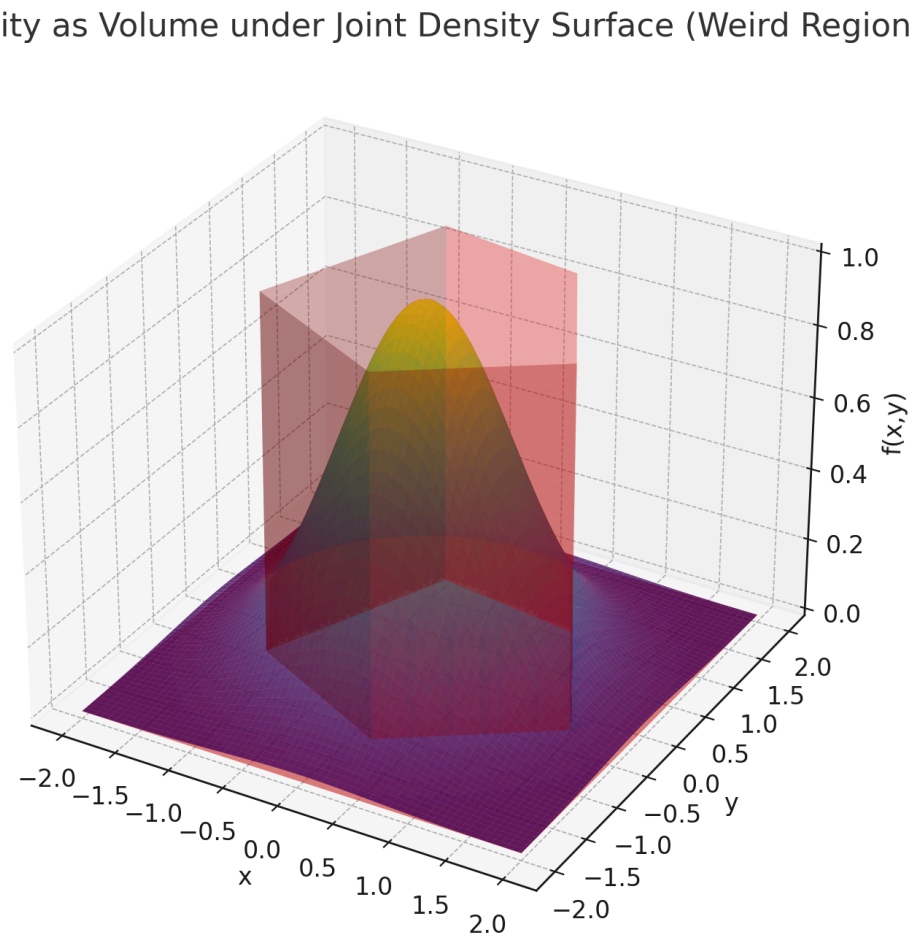
4.7 Joint Density Function

Geometrically: When X and Y are continuous random variables, their joint density function $f(x,y)$ is a surface lying above the xy plane, and $P[(X,Y) \in A]$, where A is any region in the xy plane, is equal to the volume of the right cylinder bounded by the base A and the surface.

4.8 Geometric Intuition



4.9 Geometric Intuition



4.10 Ex: Voltage Across Two Loads

Suppose two resistive loads are connected in parallel across a noisy power supply. Let X and Y denote the normalized voltages (in per-unit, so between 0 and 1) at two loads connected to the same source. The joint PDF might be

$$f_{X,Y}(x, y) = c(x + y)$$

for $0 \leq x, y \leq 1$, where c is the **normalizing constant**.

Note that this example could model two bus voltages in a noisy power system (normalized 0–1), where higher voltages tend to co-occur because of a shared source of fluctuation.

4.11 Marginal Distributions

Given joint distributions, we may be interested in knowing the **marginal distribution** of each variable, which allow you to characterize individual random variables. The **marginal distribution** of X alone and of Y alone are

$$g(x) = \sum_y f(x, y) \quad \text{and} \quad h(y) = \sum_x f(x, y)$$

for the discrete case and, for the continuous case,

$$g(x) = \int_{-\infty}^{\infty} f(x,y)\,dy \quad \text{and} \quad h(y) = \int_{-\infty}^{\infty} f(x,y)\,dx.$$

4.12 Ex: Ballpoint Pens (Again)

What are the marginal distributions of X and Y?

	$f(x,y)$	x: 0	1	2	Row Totals
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

- **The marginals can be obtained by computing column and row totals given a table like this one. For X, $g(0) = f(0,0) + f(0,1) + f(0,2) = 5/14$, $g(1) = f(1,0) + f(1,1) + f(1,2) = 15/28$, and $g(2) = f(2,0) + f(2,1) + f(2,2) = 3/28$.**

4.13 Ex: Joint Voltage (Again)

The joint distribution is

$$f_{X,Y}(x,y) = x + y, \quad 0 \le x, y \le 1,$$

so the marginal distributions for normalized voltages are

$$f_X(x) = \int_0^1 f_{X,Y}(x,y)\,dy = x + \frac{1}{2},\ 0 \le x \le 1,$$

$$f_Y(y) = \int_0^1 f_{X,Y}(x,y)\,dx = y + \frac{1}{2},\ 0 \le y \le 1.$$

4.14 Revisiting Conditional Probability

Recall the definition of conditional probability:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

provided $P(A) > 0$. Let A and B be events defined by $X = x$ and $Y = y$ for *discrete* random variables X and Y.

Then,

$$P(Y = y \mid X = x) = \frac{P(X = x, Y = y)}{P(X = x)} = \frac{f_{X,Y}(x, y)}{g_X(x)}$$

provided $g_X(x) > 0$.

4.15 Conditional Distribution

The **conditional distribution** of Y given that $X = x$ is

$$f(y \mid x) = \frac{f_{X,Y}(x, y)}{g_X(x)}$$

provided $g_X(x) > 0$. We can switch the places of X and Y.

If we want to find the probability that a random variable X falls between a and b given $Y = y$, then

$$P(a < X < b \mid Y = y) = \sum_{a<x<b} f(x \mid y) \quad \text{or} \quad \int_a^b f(x \mid y)\, dx.$$

4.16 Ex: Ballpoint Pens (Again^2)

Find the conditional distribution of X given that $Y = 1$.

	$f(x,y)$	x = 0	1	2	Row Totals
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

- **First, note that $f_Y(1) = \sum_{x=0}^{2} f(x, 1) = 3/7$. Then,**

$$f(x \mid y = 1) = \frac{f(x, 1)}{f_Y(1)} = \frac{7}{3} f(x, 1), \quad x = 0, 1, 2.$$

4.17 Ex: Ballpoint Pens (Again^2)

Now, find $P(X = 0 \mid Y = 1)$.

	$f(x,y)$	x = 0	1	2	Row Totals
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

- **To find $P(X=0 \mid Y=1)$, we use the conditional distribution:**

$$f(x=0 \mid y=1) = \frac{7}{3} f(0,1) = \frac{7}{3} \cdot \frac{3}{14} = \frac{1}{2}.$$

4.18 Ex: Joint Voltage (Again^2)

The joint and marginal distributions are

$$f_{X,Y}(x,y) = x + y, \quad 0 \le x, y \le 1,$$

$$f_X(x) = x + \frac{1}{2},\ 0 \le x \le 1, \quad f_Y(y) = y + \frac{1}{2},\ 0 \le y \le 1.$$

What is the distribution of X given that $Y = \frac{1}{3}$?

- **The conditional probability distribution is**

$$f(x \mid y = 1/3) = \frac{x + 1/3}{5/6} = \frac{6}{5}(x + 1/3), \quad 0 \le x \le 1.$$

4.19 Statistical Independence

- Before, we used conditional probabilities to determine if random variables were independent. Now, we make use of the joint distribution to do such a thing.
- Let X and Y be two random variables (discrete or continuous) with joint probability distribution $f(x,y)$ and marginal distributions $g(x)$ and $h(y)$, respectively. Then, X and Y are **statistically independent** if and only if

$$f(x,y) = g(x)h(y)$$

  for all (x,y) within their range.

4.20 Ex: Ballpoint Pens (Again^3)

Are X and Y independent?

	$f(x,y)$	x = 0	1	2	Row Totals
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

- **We can always check specific probabilities, because $f(x,y) = g(x)h(y)$ must be true for all (x,y). Let's try the point $(0,1)$. We see that $f(0,1) = \frac{3}{14}$, $g(0) = \frac{5}{14}$, and $h(1) = \frac{3}{7}$. Because $f(0,1) \neq g(0)h(1)$, then X and Y are not statistically independent.**

4.21 Ex: Joint Voltage (Again[3])

Recall that the joint distribution is

$$f_{X,Y}(x,y) = x + y, \quad 0 \leq x, y \leq 1,$$

and the marginal distributions for normalized voltages are $f_X(x) = x + \frac{1}{2},\ 0 \leq x \leq 1$ and $f_Y(y) = y + \frac{1}{2},\ 0 \leq y \leq 1$. Are X and Y independent?

- **No. Note that $g(x)h(y) = (x + \frac{1}{2})(y + \frac{1}{2}) \neq f(x,y)$. Intuitively, we wouldn't expect them to be independent because the two loads are connected to the same source!**

4.22 Extending to More Variables

- Let $f_{X_1,X_2,\ldots,X_n}(x_1, x_2, \ldots, x_n)$ be a joint distribution of the n discrete random variables $\{X_i\}_{i=1}^n$. The marginal distribution of X_j is

$$g_{X_j}(x_j) = \sum_{i \neq j} \sum_{x_i} f_{X_1,X_2,\ldots,X_n}(x_1, x_2, \ldots, x_n).$$

- The random variables are *mutually* statistically independent if

$$f_{X_1,X_2,\ldots,X_n}(x_1, x_2, \ldots, x_n) = f_{X_1}(x_1) \cdot \cdots \cdot f_{X_n}(x_n).$$

4.23 Breakout Groups

See Ch3-BO3